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Q1: Write a Java program that takes a string as input, counts how many uppercase letters, lowercase letters, and digits it contains, and creates a new string where uppercase letters are converted to lowercase and lowercase letters are converted to uppercase. Display the counts, the converted string, and the ASCII value of each character. [20 Marks]

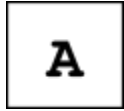
Sample Input	Sample Output
AbC123	Uppercase letters: 2 Lowercase letters: 1 Digits: 3 Converted string: aBc123 Unicode values: A -> 65 b -> 98 C -> 67 1 -> 49 2 -> 50 3 -> 51
Hello2024	Uppercase letters: 1 Lowercase letters: 4 Digits: 4 Converted string: hELLO2024 Unicode values: H -> 72 e -> 101 l -> 108 l -> 108 o -> 111 2 -> 50 0 -> 48 2 -> 50 4 -> 52

Q2: Write the Java code for this pattern [1 Mark]

Sample Input	Sample Output
4	##*#* ##*#* ##*#* ##*#*



CSE110: Programming Language I (Lab)
Spring 2026
Quiz - 02
Duration: 15 Minutes



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Q1: Write a Java program that takes a string as input, counts how many vowels, consonants, and digits it contains, and creates a new string where vowels are converted to uppercase and consonants are converted to lowercase. Display the counts, the converted string, and the ASCII value of each character. [20 Marks]

Sample Input	Sample Output
AbC123	Vowels: 1 Consonants: 2 Digits: 3 Converted string: Abc123 Unicode values: A -> 65 b -> 98 C -> 67 1 -> 49 2 -> 50 3 -> 51
Hello2024	Vowels: 2 Consonants: 3 Digits: 4 Converted string: hEllo2024 Unicode values: H -> 72 e -> 101 l -> 108 l -> 108 o -> 111 2 -> 50 0 -> 48 2 -> 50 4 -> 52

Q2: Write the Java code for this pattern [1 Mark]

Sample Input	Sample Output
4	#### #### #### ####



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Quiz - 02
Duration: 15 Minutes

B

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Q1: In a live petrol distribution system, fuel availability at 4 petrol stations is monitored as cars arrive one after another. Each station has a fixed fuel capacity, and each car requests a certain amount of fuel that must be fulfilled using the available stations.

For every car request, the system allocates fuel sequentially from Station 1 to Station 4, updating and displaying the remaining fuel after each allocation. If the system is still in the early stage and fewer than 4 allocations have been processed, it may display: "Minimum distribution not reached for stable fuel tracking".

The process continues until the user enters -1 or all petrol stations become empty. Fuel is taken in order across stations, a car may use multiple stations, and if the total fuel is insufficient for a request, the system stops and shows a failure message.

Given Array: petrol_station[] = {8, 5, 4, 9}

Sample Input	Sample Output
6 10 5 10 -1	UInitial petrol station capacities: 8 5 4 9 Car 1 (6 units request): Remaining fuel: 2 5 4 9 Car 2 (10 units request): Remaining fuel: 0 0 1 9 Car 3 (5 units request): Remaining fuel: 0 0 0 5 Car 4 (10 units request): Insufficient petrol, unable to fill the car Final petrol station capacities: 0 0 0 5



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Quiz - 03
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A

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Q2: Tracing

1	<code>public class TracingA {</code>
2	<code> public static void main(String args[]) {</code>
3	<code> int[] array = {15, 20, 10};</code>
4	<code> for (int i = 0; i < array.length; i++) {</code>
5	<code> int min_index = i;</code>
6	<code> for (int j = i + 1; j < array.length; j++) {</code>
7	<code> if (array[j] < array[min_index]) {</code>
8	<code> min_index = j;</code>
9	<code> System.out.println(min_index);</code>
10	<code> }</code>
11	<code> }</code>
12	<code> int temp = array[min_index];</code>
13	<code> array[min_index] = array[i];</code>
14	<code> array[i] = temp;</code>
15	<code> }</code>
16	<code> for (int j = 0; j < array.length; j++) {</code>
17	<code> System.out.println(array[j]);</code>
18	<code> }</code>
19	<code> }</code>
20	<code>}</code>

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Q3: Error Finding

The code below is designed for sorting an array in a descending order. However, it contains six errors. Analyze the code, identify the errors, and correct them. Provide the correction in this specific format.

1	<code>public class ErrorFindSelectionSort {</code>
2	<code> public static void main(System.in) {</code>
3	<code> int arr = {35,6,7,9,16};</code>
4	<code> for (int i = 0; i < arr.length - 1; i++) {</code>
5	<code> for (int j = 0; j < arr.length - 1; j++) {</code>
6	<code> if (arr[j] > arr[j + 1]) {</code>
7	<code> int temp = arr[j];</code>
8	<code> arr[j] = arr[j + 1];</code>
9	<code> arr[j + 1] = arr[j];</code>
10	<code> }</code>
11	<code> }</code>
12	<code> }</code>
13	<code> for (int i = 0; i < arr.length(); i++) {</code>
14	<code> System.out.print(arr[i] + " ");</code>
15	<code> }</code>
16	<code> }</code>
17	<code>}</code>

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Q2: Tracing Outputs

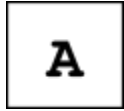
Output

Q3: Error Handling Answers

Line Number	Corrected Code
Line 2	<code>public static void main(String args[]) {</code>



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Q1: Write a Java program where an array of integers is given. Now, you have to create two new subarrays: one containing all the odd numbers and the other containing all the even numbers. Finally, print both arrays and the difference between the Largest Even Number and the Smallest Odd Number.

Sample Input	Sample Output
<code>int [] array = {5, 3, 8, 6, 7, 2, 1, 4};</code>	Odd Numbers: [5, 3, 7, 1] Even Numbers: [8, 6, 2, 4] Difference between Largest Even Number and Smallest Odd Number: 7
<code>int [] array = {2, 4, 1, 6, 12};</code>	Odd Numbers: [1] Even Numbers: [2, 4, 6, 12] Difference between Largest Even Number and Smallest Odd Number: 11



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B

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Q2: Tracing

1	public class TracingA {
2	public static void main(String args[]) {
3	int[] array = {81, 101, 11};
4	for (int i = 0; i < array.length; i++) {
5	int min_index = i;
6	for (int j = i + 1; j < array.length; j++) {
7	if (array[j] < array[min_index]) {
8	min_index = j;
9	System.out.println(min_index);
10	}
11	}
12	int temp = array[min_index];
13	array[min_index] = array[i];
14	array[i] = temp;
15	}
16	for (int j = 0; j < array.length; j++) {
17	System.out.println(array[j]);
18	}
19	}
20	}

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Q3: Error Finding

The code below is designed for sorting an array in a descending order. However, it contains six errors. Analyze the code, identify the errors, and correct them. Provide the correction in this specific format.

1	<code>public class ErrorFindSelectionSort {</code>
2	<code> public static void main(System.in) {</code>
3	<code> int arr = {35,6,7,9,16};</code>
4	<code> for (int i = 0; i < arr.length() - 1; i++) {</code>
5	<code> for (int j = 0; j < arr.length - 1; j++) {</code>
6	<code> if (arr[j] > arr[j + 1]) {</code>
7	<code> int temp = arr[j];</code>
8	<code> arr[j] = arr[j + 1];</code>
9	<code> arr[j + 1] = arr[j];</code>
10	<code> }</code>
11	<code> }</code>
12	<code> }</code>
13	<code> for (int i = 0; i < arr.length; i++) {</code>
14	<code> System.out.print(arr[i] + " ");</code>
15	<code> }</code>
16	<code> }</code>
17	<code>}</code>

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Q2: Tracing Outputs

Output

Q3: Error Handling Answers

Line Number	Corrected Code
Line 2	public static void main(String args[]) {



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B

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Q1: Write a Java program named SecondMaxPairSum that takes an array of unique integers (with at least three elements) as input, generates all possible pairs (a, b) such that a appears before b in the array, computes the sum of each pair, and then prints the second largest of these sums.

Sample Input	Sample Output	Explanation
<code>int[] arr = {2, 5, 8, 1};</code>	Second Maximum pair sum: 10	From the array {2, 5, 8, 1}, all pairs (a, b) where a comes before b are (2,5),(2,8),(2,1), (5,8),(5,1), (8,1). Their sums are: 7, 10, 3, 13, 6, 9. The second maximum pair sum among them is 10 .
<code>int[] arr = {4, 6, 3, 10, 2};</code>	Second Maximum pair sum: 14	From the array {4, 6, 3, 10, 2}, all pairs (a, b) where a comes before b are (4,6), (4,3), (4,10), (4,2), (6,3), (6,10), (6,2), (3,10), (3,2), (10,2). Their sums are: 10, 7, 14, 6, 9, 16, 8, 13, 5, 12. The second maximum pair sum among them is 14 .



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Q2: Tracing

1	public class TracingA {
2	public static void main(String args[]) {
3	int[] array = {20, 10, 5};
4	for (int i = 0; i < array.length; i++) {
5	int min_index = i;
6	for (int j = i + 1; j < array.length; j++) {
7	if (array[j] < array[min_index]) {
8	min_index = j;
9	System.out.println(min_index);
10	}
11	}
12	int temp = array[min_index];
13	array[min_index] = array[i];
14	array[i] = temp;
15	}
16	for (int j = 0; j < array.length; j++) {
17	System.out.println(array[j]);
18	}
19	}
20	}

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Q3: Error Finding

The code below is designed for sorting an array in a descending order. However, it contains six errors. Analyze the code, identify the errors, and correct them. Provide the correction in this specific format.

1	<code>public class ErrorFindSelectionSort {</code>
2	<code> public static void main(System.in) {</code>
3	<code> int arr = {35,6,7,9,16}</code>
4	<code> for (int i = 0; i < arr.length - 1; i++) {</code>
5	<code> for (int j = 0; j < arr.length - 1; j++) {</code>
6	<code> if (arr[j] > arr[j + 1]) {</code>
7	<code> int temp = arr[j];</code>
8	<code> arr[j] = arr[j + 1];</code>
9	<code> arr[j + 1] = arr[j];</code>
10	<code> }</code>
11	<code> }</code>
12	<code> }</code>
13	<code> for (int i = 0; i < arr.length; i++) {</code>
14	<code> System.out.print(arr[i] + " ");</code>
15	<code> }</code>
16	<code> }</code>
17	<code>}</code>

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Q2: Tracing Outputs

Output

Q3: Error Handling Answers

Line Number	Corrected Code
Line 2	public static void main(String args[]) {



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```
public class quiz04 {
    public static void main(String[] args) {
        MethodA();
        String s1 = "";
        int [] array = {104, 111, 110, 111, 114, 32, 116, 111, 32, 115, 101, 114, 118, 101};
        String s3 = MethodC(array,s1);
        System.out.print(s3);
        String st = " you ";
        String final_string = MethodB(st);
        System.out.print(final_string);
    }
    public static String MethodB(String sp1){
        char s5 = (char) ((sp1.charAt(0) - 1) * 0 + 97);
        String s0 = sp1 + s5+ MethodD() +",Thank" + sp1 + '!!';
        return s0;
    }
    public static String MethodC(int [] a, String s){
        for (int i = 0; i < a.length;i++){
            s += (char) (a[i]);
        }
        return s;
    }
    public static String MethodD(){
        return "ll";
    }
    public static void MethodA(){
        int a = 5;
        int b = ++a;
        int sum = a++ + ++b;
        if (sum == 12){
            System.out.print("CSE is Fun");
        }
        else if (sum == 13){
            System.out.print("It was a great ");
        }
        else if (sum == 14){
            System.out.print("Last Quiz ");
        }
    }
}
```